INCH-POUND

MIL-T-50036B
24 December 1992

SUPERSEDING
MIL-T-50036A
22 July 1980

MILITARY SPECIFICATION

TALC, TECHNICAL, T1 AND T3

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of technical grade powdered talc.

1.2 Classification. Talc, technical shall be of the following types as specified (see 6.2):

Type T1
Type T3

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-PET-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810

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SPECIFICATIONS

FEDERAL

PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-186	-	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals
PPP-F-320	-	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-T-66	-	Tape, Packaging, Vinyl Plastic Film

MILITARY

MIL-B-117	-	Bags, Sleeves and Tubing
MIL-T-22085	-	Tapes, Pressure-Sensitive, Adhesive, Preservation and Sealing

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1233	-	Procedures for Determining Particle Size, Particle Size Distribution, and Packed Density of Powdered Material

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

PURCHASE DESCRIPTIONS

196-21-87	-	Bottle, Screw-Cap, Plastic, 3-1/2 Gallon

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(Copies are available from Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-PET-O, Aberdeen Proving Ground, MD 21010-5423.)

U.S. ENVIRONMENTAL PROTECTION AGENCY, RESEARCH AND DEVELOPMENT

TEST METHODS

EPA-600/M4-82-020 - Interim Method for the Determination of Asbestos in Bulk Insulation Samples

(Copies are available from TSCA Assistance Information Service, U.S. EPA, Mail Stop TS-799, Washington, D.C. 20460.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

ASTM STANDARDS

D 185 - Coarse Particles in Pigments, Pastes, and Paints
D 717 - Analysis of Magnesium Silicate Pigment
D 1208 - Common Properties of Certain Pigments
D 1895 - Apparent Density, Bulk Factor, and Pourability of Plastic Materials
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103-1187.)

National Motor Freight Traffic Association, Inc., Agent

"National Motor Freight Classification"

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

Uniform Classification Committee, Agent

"Uniform Freight Classification"

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Chemical and physical characteristics. Talc, technical shall conform to the applicable chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Type T1		Type T3		Test paragraph
	Minimum	Maximum	Minimum	Maximum	
Silicon dioxide, percent by weight	35	65	55	65	4.2.4.1
Calcium oxide, percent by weight	---	---	---	0.30	4.2.4.1
Magnesium oxide, percent by weight	20	40	25	35	4.2.4.1
Loss on ignition, percent by weight	---	---	---	8	4.2.4.2
Asbestos and other fibers	---	None detected	---	None detected	4.2.4.3
Percent by weight retained on a No. 400 sieve	---	5	---	---	4.2.4.4
Average particle size, micrometers	---	---	---	3.5	4.2.4.5
Apparent density, grams per cubic centimeter	---	---	---	0.25	4.2.4.6

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Contractor assurance of compliance. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing, special packaging instruction, and specification requirements using, as a minimum, the conformance criteria specified herein.

4.1.3 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, and other types of sampling procedures may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the inspection provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval. (See 6.3.) When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the quality assurance provisions specified herein. In cases of dispute as to whether the contractor's proposed alternative provides equal quality assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

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4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the talc, technical of one type, produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.4).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with table II. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment.

TABLE II. Sampling for examination of packaging

Number of containers in batch or lot	Number of sample containers
1	1
2 to 15	2
16 to 50	3
51 to 150	5
151 to 500	8
501 to 3,200	13
3,201 to 35,000	20
35,001 to 500,000	32
Over 500,000	50

4.2.2.2 For test. Sampling shall be conducted in accordance with table III. A representative specimen of approximately 450 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

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TABLE III. Sampling for test

Number of containers in batch or lot	Number of sample containers
1	1
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. Sample unit and shipping containers shall be examined for the characteristics listed below. Failure of any sample unit or shipping container to conform to all characteristics shall be cause for rejection of the lot represented.

(a) Unit pack containers and closures correct
(b) Packing container and closure correct
(c) Unit pack and packing container not damaged nor leaking
(d) Tape correct (where required)
(e) Fiberboard pads correct (where required)
(f) Strapping correct (where required)
(g) Marking correct

4.2.3.2 For test. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 450 g. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. Tests shall be conducted as follows:

4.2.4.1 Silicon dioxide, calcium oxide, and magnesium oxide. Determine the percent by weight silicon dioxide, calcium oxide, and magnesium oxide in the specimen in accordance with the applicable procedures in ASTM D 717.

4.2.4.2 Loss on ignition. Determine the percent by weight loss on ignition of the specimen in accordance with the procedure for Loss on Ignition and Ash in ASTM D 1208.

4.2.4.3 Asbestos and other fibers. Talc, technical shall be fiber free. For purposes of this test a fiber is a particle that has a length greater than five micrometers and an aspect

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ratio (length to width ratio) of three or more to one. Determine the presence of fibers in accordance with test method EPA-600/M4-82-020. When authorized by the procuring activity, this method may be supplemented by procedures using X-ray diffraction, transmission electron microscopy with selected area electron diffraction, and energy dispersive X-ray.

4.2.4.4 Percent retained on a No. 400 sieve. Determine the percent by weight of specimen retained on a No. 400 sieve in accordance with the procedure for Insoluble Dry Pigments, Except Metallic Aluminum and Bronze Powders in ASTM D 185, except use a No. 400 sieve instead of a No. 325 sieve.

4.2.4.5 Average particle size. Determine the average particle size of the specimen in accordance with method 100 of MIL-STD-1233.

4.2.4.6 Apparent density. Determine the apparent density of the specimen in accordance with method A of ASTM D 1895.

5. PACKAGING

Note: The metric equivalents given for inch-pound quantities are nominal values provided for informational purposes and should not be considered as quantity requirements.

5.1 Preservation. Talc shall be preserved level A, level C, or commercial as specified (see 6.2).

5.1.1 Level A unit packing. Talc shall be unit packed in 5-, 8-, or 29-pound (lb) [2.3-, 3.6-, or 13.2-kilogram (kg)] quantities as specified (see 6.2).

5.1.1.1 Five-lb (2.3-kg) quantity of type T1 talc (for Navy use). A quantity of 5 lb (± 2 ounces) (2.3 kg) of type T1 talc, shall be unit packed in a plastic container conforming to class 2, grade 1 or 2, closure B, style optional of PPP-C-186. The metal screw-cap shall be provided with an interior liner that is compatible with the talc. After proper closure is completed, the screw-cap shall be secured with a vinyl polyester pressure-sensitive sealing tape conforming to type 1, class optional of PPP-T-66 or a self-shrinking band to prevent loosening of the screw-cap during handling and shipping. The tape shall have width as required and a length of not less than one and one-third times the circumference of the screw-cap closure. The plastic container shall be placed within a fiberboard box conforming to class domestic of PPP-B-636. All other box options such as variety, grade, and style shall be at the contractor's discretion. The box shall be closed by the use of pressure-sensitive tape in accordance with Method I in the appendix to PPP-B-636.

5.1.1.2 Twenty-nine-lb (13.2-kg) quantity of type T1 talc. A quantity of 29 lb (± 8 ounces) (13.2 kg) of type T1 talc shall be unit packed in a 14 by 24 inch [356 by 610 millimeter (mm)] notched heat-sealable bag conforming to type I, class E, style 1 of MIL-B-117. The filled and sealed bag shall be placed inside a paper sack which shall comply

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with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other carrier rules as applicable to the mode of transportation. Care shall be exercised to avoid puncturing the interior bag during the exterior bag operation.

5.1.1.3 Eight-lb (3.6-kg) quantity of type T3 talc. A quantity fo 8 lb (± 2 ounces) (3.6 kg) of type T3 talc shall be unit packed in a screw-cap plastic bottle conforming to Purchase Description 196-21-87. The tightened screw-cap closure shall be further secured to the neck of the bottle by 2-inch (51-mm) wide tape conforming to type II of MIL-T-22085. The tape shall be wound around the neck of the bottle and the cap a minimum of two complete turns. The tape shall extend from the top of the cap to a circumferential line around the bottle spout no less than 3/4 inch (19 mm) below the bottom of the tightened cap.

5.1.2 Level C (for Navy use). A quantity of 5 lb (± 2 ounces) (2.3 kg) of type T1 talc shall be unit packed as specified in 5.1.1.1 except that the fiberboard box shall conform to the applicable carrier rules and regulation.

5.1.3 Commercial unit packing. A quantity of talc as specified (see 6.2) shall be unit packed in accordance with ASTM D 3951.

5.2 Packing. Talc shall be packed level A, level B, level C, or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Five-lb (2.3-kg) quantity of type T1 talc (for Navy use). Six 5-lb (2.3-kg) packs of type T1 talc prepared as specified in 5.1.1.1 or 5.1.2 shall be packed in a snug-fitting fiberboard box conforming to type CF, class weather-resistant of PPP-B-636. The box variety grade and style shall be at the contractor's discretion. The box shall be closed and waterproofed in accordance with method V and reinforced with pressure-sensitive, filament-reinforced tape in accordance with the requirements specified in the appendix to PPP-B-636.

5.2.1.2 Twenty-nine-lb (13.2-kg) quantity of type T1 talc. Twelve 29-lb (13.2-kg) unit packs of type T1 talc prepared as specified in 5.1.1.2 shall be packed in a 55-gallon (208-liter), fully removable head steel conforming to Uniform Freight Classification Rule 40. The interior arrangement shall be two tiers with six sacks per tier. The tiers shall be separated by a fiberboard pad fabricated from material conforming to class weather-resistant, variety SW, grade W5c or W6c of PPP-F-320. The box shall be closed and strapped in accordance with the closure and strapping requirements specified in the appendix to the applicable box specification.

5.2.1.3 Eight-lb (3.6-kg) quantity of type T3 talc. Five 8-lb (3.6-kg) unit packs of type T3 talc prepared as specified in 5.1.1.3 shall be packed side by side in an upright position, with pouring spouts facing in alternate directions, in a wood box conforming to grade A,

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class 2, style 4-1/2 of PPP-B-621 or grade A, type overseas, style I of J of PPP-B-601. The interior surfaces of the box shall be completely lined with fiberboard pads. A flat fiberboard pad extending the full length of the box shall be placed above the handles and between the spouts of the plastic bottles. The pad shall extend 1/4 ± 1/8 inch (6.4 mm) above the tops of the spouts. Additional fiberboard pads shall be used where needed to assure a tight pack. All fiberboard pads shall be fabricated from material conforming to class weather-resistant, variety SW, grade W5c of PPP-F-320. The box shall be closed and strapped in accordance with the closure and strapping requirements specified in the appendix to the applicable box specification.

5.2.2 Level B.

5.2.2.1 Five-lb (2.3-kg) quantity of type T1 talc (for Navy use). Six 5-lb (2.3-kg) unit packs of type T1 talc prepared as specified in 5.1.1.1 or 5.1.2 shall be packed as specified in 5.2.1.1 except that the fiberboard box shall conform to class domestic of PPP-B-636. The box shall be closed by the use of pressure-sensitive tape in accordance with Method I in the appendix to PPP-B-636.

5.2.2.2 Eight-lb (3.6-kg) quantity of type T3 talc. Five eight-lb (3.6-kg) unit packs of type T3 talc prepared as specified in 5.1.1.3 shall be packed as specified in 5.2.1.3 except that the wood box shall conform to class 1, style 4-1/2 of PPP-B-621; type domestic, style I or J of PPP-B-601; or class 2 of PPP-B-585. Wood preservative is not required.

5.2.3 Level C (for Navy use). Six 5-lb (2.3-kg) unit packs of type T1 talc prepared as specified in 5.1.1.1 or 5.1.2 shall be packed in containers acceptable to the common carrier and which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing, and method of shipment shall comply with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other carrier rules as applicable to the mode of transportation.

5.2.4 Commercial. Talc shall be commercially packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Levels A, B, and C. In addition to the special marking required herein, or by the contract or order (see 6.2), unit containers, including both the plastic and fiberboard containers specified in 5.1.1.1, and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial interior packs and exterior shipping containers shall be marked in accordance with ASTM D 3951.

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5.3.3 Special marking. Each interior pack and exterior shipping container shall be marked as follows: "TALC - ASBESTOS-FREE."

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Types T1 and T3 talc are intended for use as simulants for Riot Control Agents CS, CS1, and CS2. Type T1 is also intended for use in dusting and lubrication.

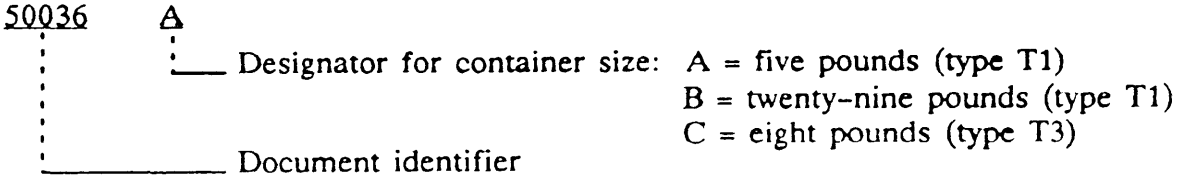
6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see section 2)
(c) Type of talc required (see 1.2)
(d) Unit quantity required (see 5.1.1, 5.1.2, and 5.1.3)
(e) Level of preservation required (see 5.1)
(f) Level of packing required (see 5.2)
(g) Special marking, if required (see 5.3.1)

6.3 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.4 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.5 Part identification numbering system. A recommended system is as follows:



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6.6 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Subject term (key word) listing.

asbestos
riot control agents

Custodian:

Army - EA
Navy - SH

Review activities:

Army - MD
Air Force - 68
DLA - GS

User activity:

Navy - OS

Preparing activity:

Army - EA

Project No. 6810-1284

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-50036B	2. DOCUMENT DATE (YYMMDD) 921224

3. DOCUMENT TITLE
TALC, TECHNICAL, T1 AND T3

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
U.S. Army Edgewood Research, Development and Engineering Center	(410) 671-3259	DSN 584-3259
c. ADDRESS (Include Zip Code) Tech Dir, U.S. Army ERDEC ATTN: SCBRD-ENE-S (Std/Specs/Pkg) Aberdeen Proving Ground, MD 21010-5423	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 Previous edition is obsolete 198/290

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